Wed. 2 Apr 2025

Last time we derived Newton's method

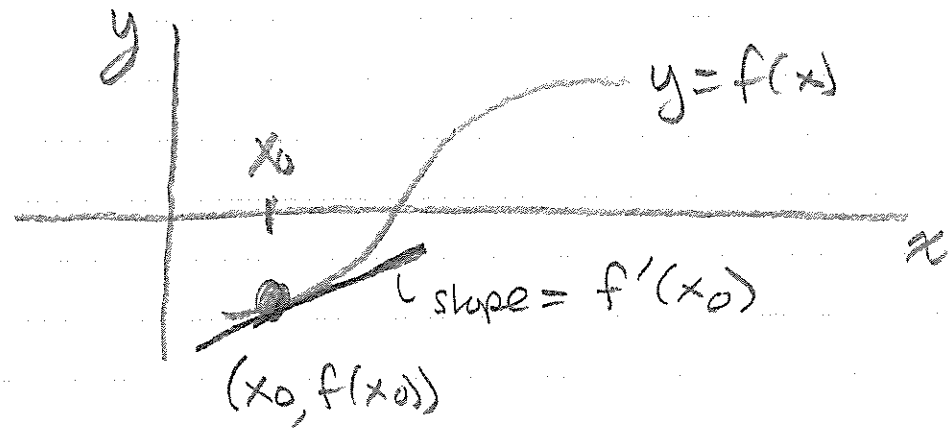
$$\begin{cases} x_0 \text{ given} \\ x_{i+1} = x_i - \dfrac{f(x_i)}{f'(x_i)} \qquad i = 0, 1, 2, \ldots \end{cases}$$

for finding a root (or trying to) of $f(x)$.

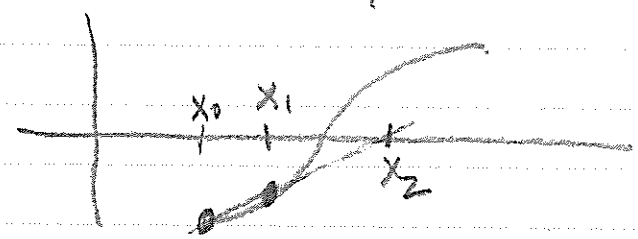
We observed that it converged very quickly for our example problem.

Finding derivatives can be an error prone process, so we'll look for simpler processes.

To derive Newton's method we considered a linear approximation to $y = f(x)$ near x_0 that had slope $f'(x_0)$



Suppose instead we picked some point x_1



and determine the line that interpolates $(x_0, f(x_0))$, $(x_1, f(x_1))$.
It is called a secant line (a straight line that cuts a curve at two points.)

Its equation is $y - f(x_1) = \dfrac{f(x_1) - f(x_0)}{x_1 - x_0} \cdot (x - x_1)$

and its x-intercept is $x = x_1 - \dfrac{f(x_1)(x_1 - x_0)}{f(x_1) - f(x_0)}$

This suggests taking the next approximation to x^* as

$$x_2 = x_1 - \frac{f(x_1)(x_1 - x_0)}{f(x_1) - f(x_0)}$$

note: does not require knowledge of $f'(x)$

and defining the iteration scheme

$$\begin{cases} x_{i+1} = x_i - \dfrac{f(x_i)(x_i - x_{i-1})}{f(x_i) - f(x_{i-1})}, & i = 1, 2, 3, \ldots \\ \text{where } x_0, x_1 \text{ given} \end{cases}$$

This method is called the <u>Secant Method</u>.

Note: only requires <u>one</u> new function evaluation per iteration while Newton's method required <u>two</u> $(f(x_i), f'(x_i))$.

(So less "work" to compute per iteration)

<u>example</u> Use the Secant Method to compute a root of
$f(x) = x - 0.2\sin(x) - 0.5$
starting with $x_0 = 0$, $x_1 = 1$ and
stop when $|x_i - x_{i-1}| < 5 \times 10^{-7}$.

We can compute the following results

i	x_i	$\Delta x_i = \lvert x_i - x_{i-1} \rvert$
0	0	
1	1	
2	0.6011741...	4.0×10^{-1}
3	0.6150404...	1.4×10^{-2}
4	0.6154686...	4.3×10^{-4}
5	0.6154682...	4.2×10^{-7}

We meet the stopping criteria in 4 steps, which is the same as Newton's method (though Newton's method has a much smaller Δx_i by the 4th iteration)

So the secant method can converge very quickly.

I had said that the bisection method converges <u>slowly</u>, and that Newton's method and the secant method converge <u>quickly</u>. (when they converge)

How can we quantify this concept and be more precise?

Rates of Convergence

Suppose we generate a sequence

$$x_0, x_1, x_2, \ldots$$

that converges to x^*.

Let $e_i = |x_i - x^*|$ be the absolute error in term x_i. (Sometimes we'll take e_i to be a bound on this error)

The <u>rate of convergence</u> r of the sequence is the largest value r such that

$$\lim_{\substack{x_i \to x^* \\ (e_i \to 0)}} \frac{e_{i+1}}{(e_i)^r} = c \neq 0$$

where c is some nonzero constant.

For bisection, we have seen that

$$e_{i+1} = \tfrac{1}{2} e_i$$

and so $\displaystyle\lim_{x_i \to x^*} \frac{e_{i+1}}{e_i^1} = 1/2$ and $r = 1$.

When $r = 1$, we say that the method has <u>linear convergence</u>.

When $r > 1$, the method is said to have <u>superlinear</u> convergence

What is the convergence rate for Newton's Method?

Consider a Taylor series for $f(x)$ about $x = x_i$

We know
$$\begin{aligned} 0 &= f(x^*) \\ &= f(x_i + (x^* - x_i)) \\ &= f(x_i) + f'(x_i)(x^* - x_i) + \tfrac{1}{2} f''(\theta)(x^* - x_i)^2 \end{aligned}$$

for some θ between x_i, x^*

$\div$ both sides by $f'(x_i)$ [assume it exists and is nonzero] gives

$$0 = \frac{f(x_i)}{f'(x_i)} + (x^* - x_i) + \frac{1}{2}\frac{f''(\theta)}{f'(x_i)}(x^* - x_i)^2$$

rearrange:
$$x^* - \left(x_i - \frac{f(x_i)}{f'(x_i)}\right) = -\frac{1}{2}\frac{f''(\theta)}{f'(x_i)}(x^* - x_i)^2$$

or
$$x^* - x_{i+1} = -\frac{f''(\theta)}{2f'(x_i)}(x^* - x_i)^2$$

and so
$$e_{i+1} = \left|\frac{f''(\theta)}{2f'(x_i)}\right| e_i^2$$

$\therefore$ if the method converges,

$$\lim_{x_i \to x^*} \frac{e_{i+1}}{e_i^2} = \left|\frac{f''(x^*)}{2f'(x^*)}\right| = \text{constant}$$

so $r = 2$, and we say Newton's method has quadratic convergence.

[If $f'(x^*) = 0$, x^* is a multiple root and it turns out convergence

"mean" is

<u>Secant rule</u> : You can show that

$$\lim_{x_i \to x^*} \frac{e_{i+1}}{e_i e_{i-1}} = c \neq 0$$

and $$\lim_{x_i \to x^*} \frac{e_{i+1}}{e_i^r} = c \neq 0 \text{ for } r = \frac{1+\sqrt{5}}{2} \doteq 1.618$$

How can we interpret these convergence rates?

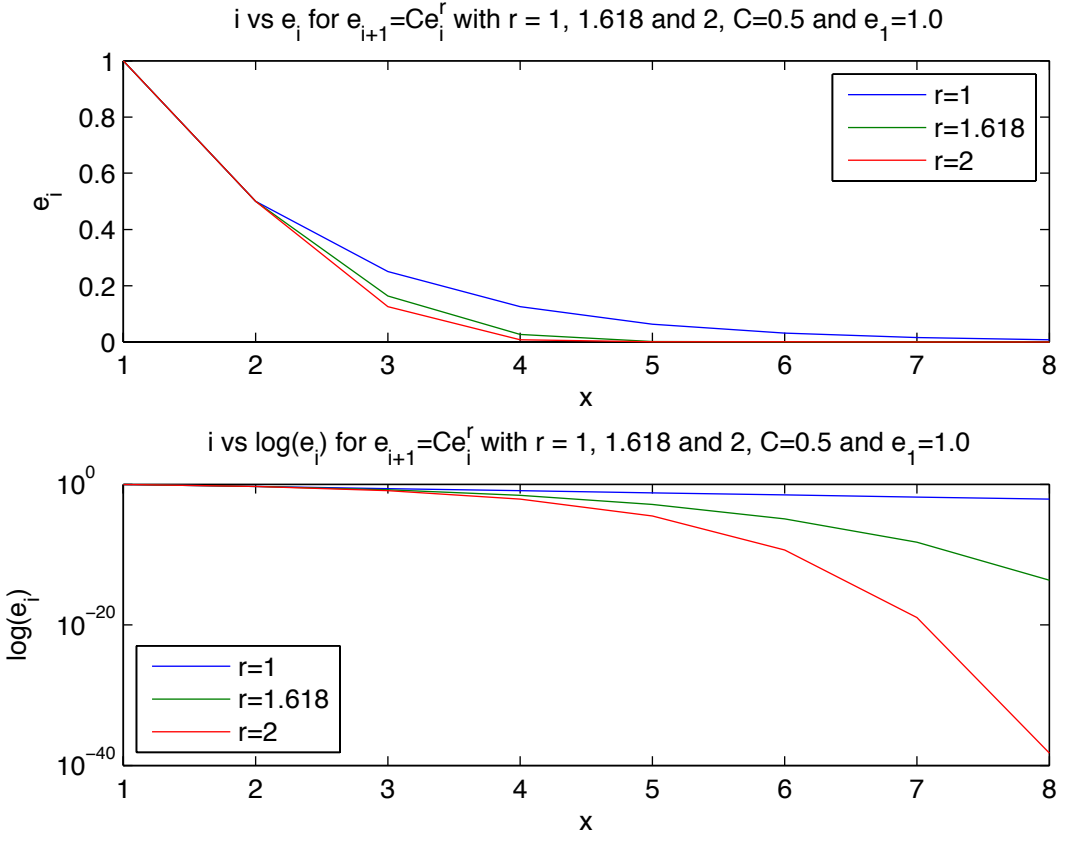
Suppose we have the sequences $e_1 = 1$, $e_{i+1} = \frac{1}{2} e_i^r$, $i = 0, 1, 2, \ldots$ with $r = 1$ or 1.618 or 2.

Graphs of these three sequences are in the top figure.

Note:
- each sequence converges toward 0 and the $r=2$ iteration gets there first.
- the difference in the speed of convergence is more noticeable in the graph of i vs $\log(e_i)$
  The $r=1$ curve is a straight line.
  The superlinear methods decay rapidly and go towards 0 faster.

for $r=1$, $e_8 = 7.8 \times 10^{-3}$
$r=1.618$, $e_8 = 2.21 \times 10^{-14}$
$r=2$, $e_8 = 5.88 \times 10^{-39}$

$\therefore$ want to use a superlinear method when possible as they converge to root much faster.


i vs e_i for $e_{i+1}=Ce_i^r$ with r = 1, 1.618 and 2, C=0.5 and e_1=1.0
e_i
x
r=1
r=1.618
r=2
i vs log(e_i) for $e_{i+1}=Ce_i^r$ with r = 1, 1.618 and 2, C=0.5 and e_1=1.0
log(e_i)
x
r=1
r=1.618
r=2